Lecture 7&8 – Lab 7&8

Sorting Algorithms: Selection, Insertion, Merge, Quick

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Goal: Sort the array `[3, 9, 1, 7, 2]`

In insertion sort algorithm we make **comparison and shift** numbers if comparison succeed.
Apply insertion sort algorithm to sort this array,
and write how many comparison & shift occurred to sort it?

Based on the following array, apply the selection sort and state how many swap operation executed?

Answer the following questions

- Worst case of Insertion Sort = ……
- Best case of Insertion Sort = ……
- Average case of Insertion Sort = ……
- Space complexity for Insertion Sort = ……

- Worst case of Selection Sort = ……
- Best case of Selection Sort = ……
- Average case of Selection Sort = ……
- Space complexity for Selection Sort = ……

- Insertion sort is: (a) in-place (b) out-of-place
- Selection sort is: (a) in-place (b) out-of-place
- Insertion sort is: (a) stable (b) not stable
- Selection sort is: (a) stable (b) not stable

Answer the following questions

- Worst case of Merge Sort = ……
- Best case of Merge Sort = ……
- Average case of Merge Sort = ……
- Space complexity for Merge Sort = ……

- Worst case of Quick Sort = ……
- Best case of Quick Sort = ……
- Average case of Quick Sort = ……
- Space complexity for Quick Sort = ……

- Merge sort is: (a) in-place (b) out-of-place
- Quick sort is: (a) in-place (b) out-of-place
- Merge sort is: (a) stable (b) not stable
- Quick sort is: (a) stable (b) not stable

Write and implement the following Algorithms

- Insertion sort
- Selection sort
- Merge sort
- Quick sort

1. **[Your Task]** Prepare presentation for the following sorting algorithms, including (idea, example, algorithm, time and space complexity, when to use this algorithm):
   - **Heap sorting**
   - **Count sort**
   - **Radex sort**